CLARIANT

Clariant Produkte (Deutschland) GmbH
Analytical Sciences, G860
Dr. Viktor Heinrichs

REPORT OF ANALYSIS - PHYSICS, PARTICLE & SURFACE ANALYSIS

Sample:	**MB3001A, MB3004A and MB3001A/MB3004A blend**
Order title/Product:	Graphene / Raman microscopy
Project:	Graphene
Account:	Curia Germany GmbH
Customer:	Dr. Thorsten Busch
Order no.:	24-048758
Remarks:	Raman measurement on request of the customer

Method:

Raman spectroscopy/microscopy using Senterra I Raman microscope (Bruker).

Procedure:

Objective:	Raman spectroscopy
Field of Application:	Powder
Principle:	The substance is applied to a microscope slide and examined under the microscope. The objective is selected and focused to examine an exact location using Raman. At this position, several points are examined with the laser and a corresponding number of Raman spectra are recorded.
Reference:	

Experimental Details and Result

Equipment:	Senterra I Raman microscope (Bruker)
Magnification:	50 x
Number of spectra per position	12
Evaluation:	Averaging over several spectra

Result:

Spectra see attachment.

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24-048758-1: MB3001A

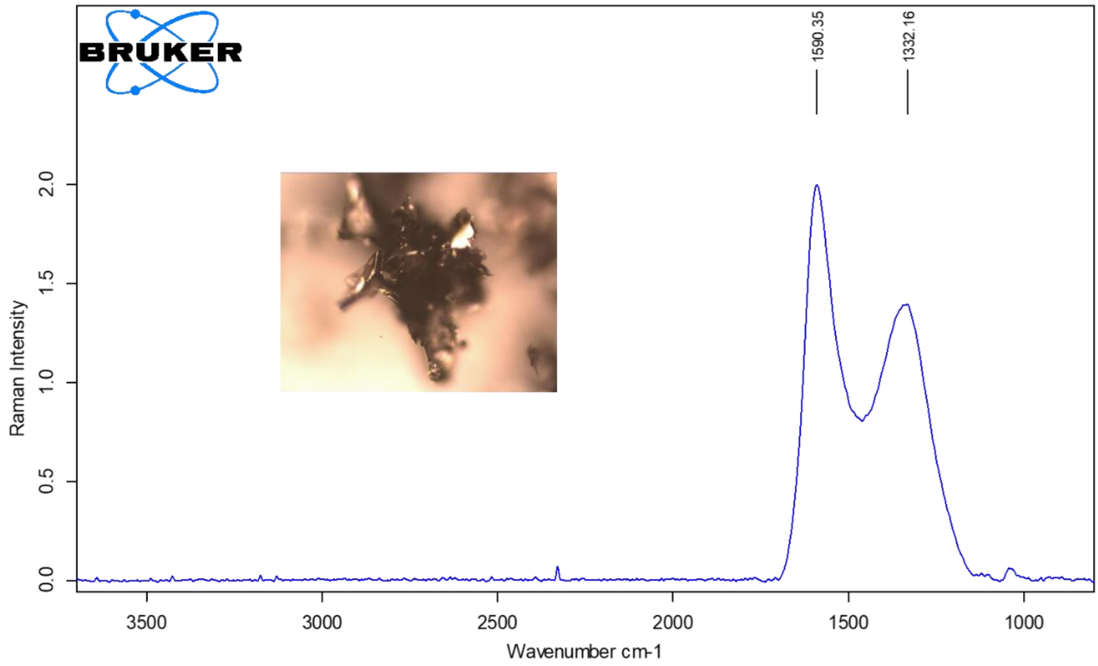

Figure 1: MB3001A, blue curve: average spectrum over 12 recordings

24-048758-2: MB3004A

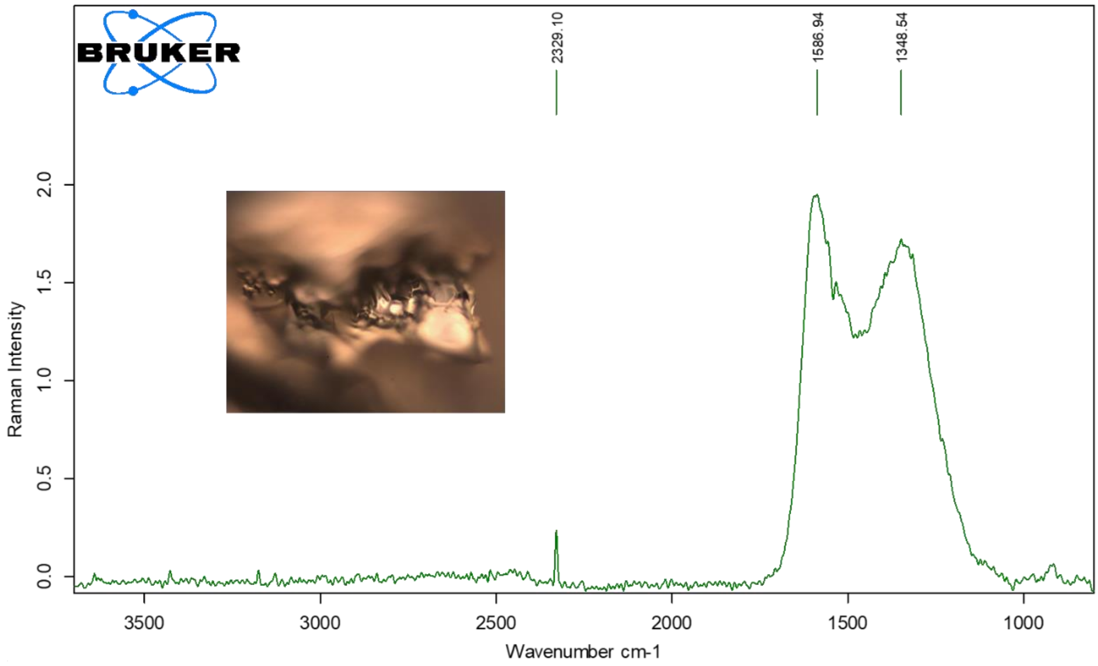

Figure 2: MB3004A, green curve: average spectrum over 12 recordings

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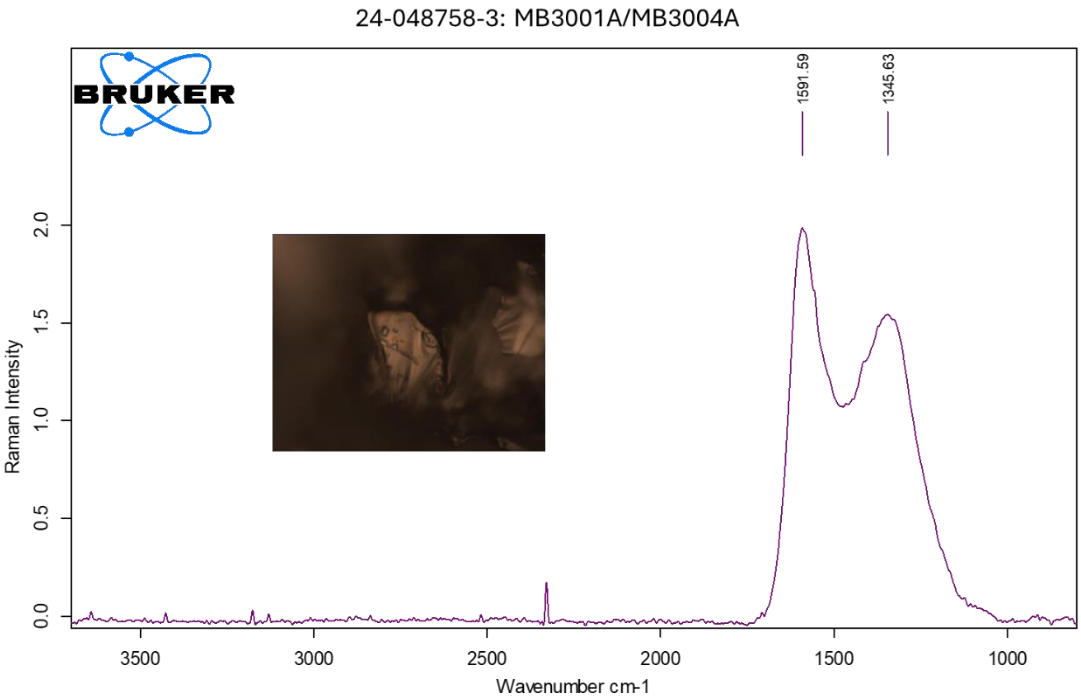

Figure 3: MB3001A/MB3004A Blend, violet curve: average spectrum over 12 recordings

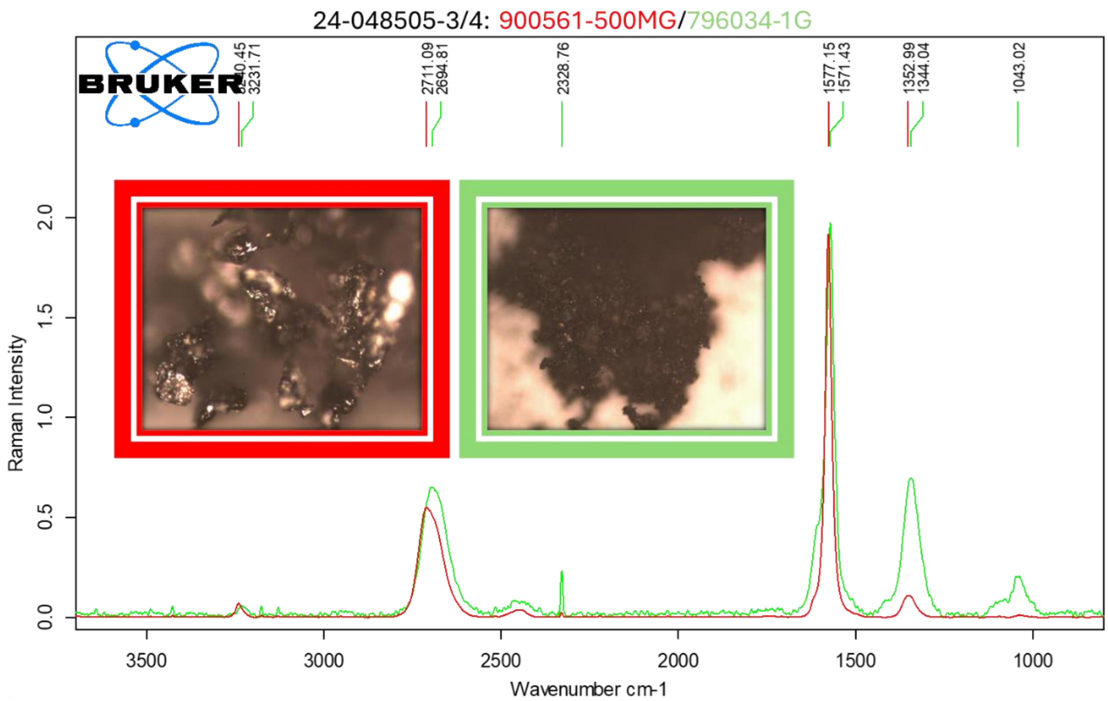

Figure 4: Graphen 900561-500MG (red curve) and Graphene oxid 796034-1G (green curve): average spectrum over 12 recordings

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